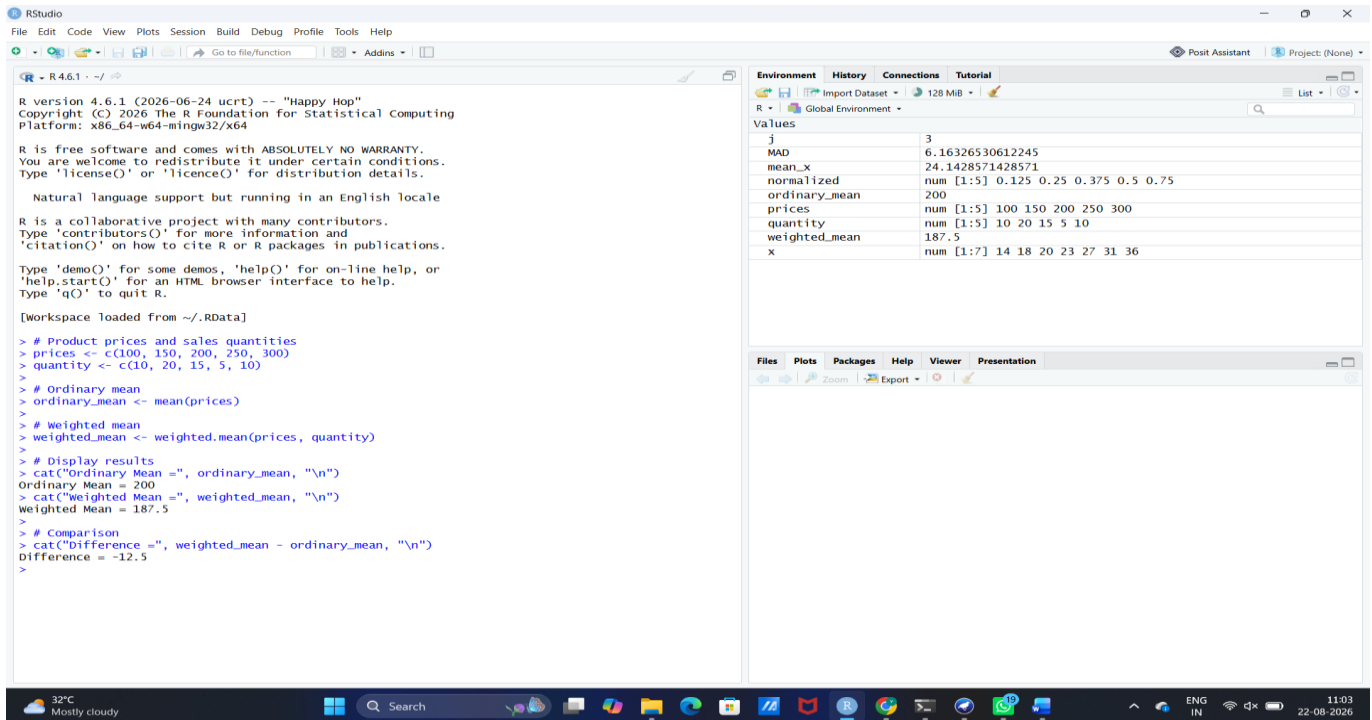
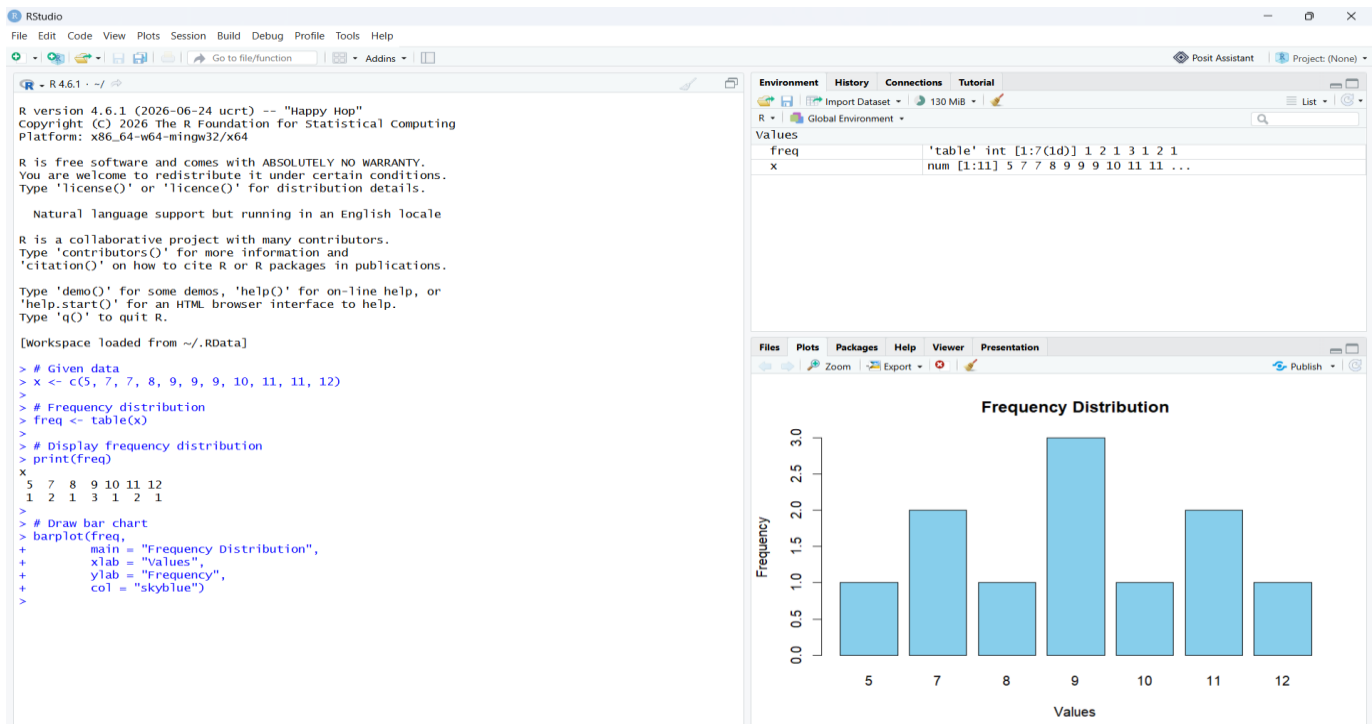


SET 11

1. R: Calculate weighted mean for five products using prices and sales quantities; compare with ordinary mean.



2. R: Build a frequency distribution for 5,7,7,8,9,9,9,10,11,11,12 and draw a bar chart.



3. WEKA: Create email ARFF data (Length, ContainsLink, ContainsAttachment, Class); use Naive Bayes to classify a test instance.

Dataset: Email Spam

```
@relation email_spam
@attribute Length {Short,Long}
@attribute ContainsLink {Yes,No}
@attribute ContainsAttachment {Yes,No}
@attribute Class {Spam,NotSpam}
@data
Short,Yes,No,Spam
Long,Yes,Yes,Spam
Short,No,No,NotSpam
Long,No,Yes,NotSpam
Long,Yes,No,Spam
Short,No,Yes,NotSpam
Long,Yes,Yes,Spam
Short,Yes,No,Spam
```

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'NaiveBayes'. Under 'Test options', 'Use training set' is selected. The 'Classifier output' pane displays the following results:

Time taken to build model: 0 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	8	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0.1437		
Root mean squared error	0.1507		
Relative absolute error	30.2494	%	
Root relative squared error	31.0811	%	
Total Number of Instances	8		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Spam
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	NotSpam
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

```
a b <-- classified as
5 0 | a = Spam
0 3 | b = NotSpam
```

The 'Result list' on the left shows a single entry: '11:08:25 - bayes.NaiveBayes'.

4. WEKA: Apply J48 to the email data and compare accuracy, confusion matrix and F1 with Naive Bayes.

Dataset: Email Spam – J48

Use the same Email Spam ARFF dataset from Set 11 Q3.

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Classifier

ChooseJ48 -C 0.25 -M 2

Test options

☒ Use training set

☐ Supplied test set

Set...

☐ Cross-validation

Folds10

☐ Percentage split

%66

More options...

(Nom) Class

StartStop

Result list (right-click for options)

11:08:25 - bayes.NaiveBayes11:10:00 - trees.J48

Classifier output

Time taken to build model: 0.01 seconds

=== Evaluation on training set ===

Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances	8	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	8		

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Spam
	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	NotSpam
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	

=== Confusion Matrix ===

a b <-- classified as

5 0 | a = Spam

0 3 | b = NotSpam

Status

OK

Log

x 0